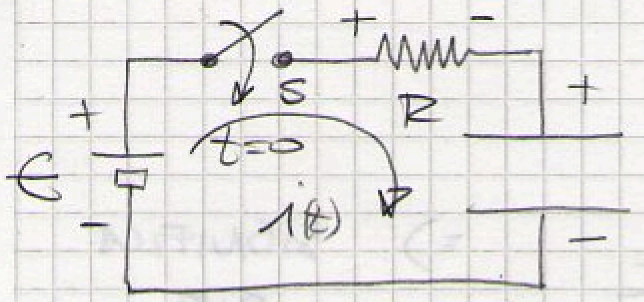
RESOLUCIÓN DETALLADA DEL CIRCUITO "R-C" =

A PARTIR DE LA ECUACIÓN DE AMPÈRE-MAXWELL:



$$+\epsilon - i(t)\cdot R - \frac{q(t)}{C} = 0$$

PERO COMO $i(t) = \frac{dq(t)}{dt} \Rightarrow$

$$\epsilon - \frac{dq(t)}{dt}\cdot R - \frac{q(t)}{C} = 0 \Rightarrow \text{RE-ORDENANDO:}$$

$$\boxed{\frac{dq(t)}{dt} + \frac{q(t)}{RC} - \frac{\epsilon}{R} = 0}$$ EC. DIFERENCIAL DE PRIMER ORDEN.

RESOLUCIÓN = VOLVIENDO A RE-ACOMODAR:

$$\frac{dq}{dt} = \left(\epsilon\cdot C - q\right)\cdot\frac{1}{RC} \Rightarrow$$

$$\frac{dq}{\epsilon\cdot C - q} = \frac{dt}{RC} \Rightarrow \int_{q=0}^{q=Q_{final}} \frac{dq}{\epsilon C - q} = \int_0^{t_f} \frac{dt}{RC}$$

CAMBIO DE VARIABLES =

$$\mu = \epsilon\cdot C - q(t) \Rightarrow \boxed{d\mu = -dq}$$ Y AHORA

SUS LÍMITES SERÁN = $\begin{cases} \text{EN } t=0;\ q(0)=0 \Rightarrow \mu_i = \epsilon C \\ \text{EN } t=t_f;\ q(t_f)=q(t) \Rightarrow \mu_f = \epsilon\cdot C - q(t) \end{cases}$

ASÍ QUE:

$$\int_{\epsilon \cdot C}^{\epsilon \cdot C - q(t)} \frac{du}{u} = -\int_0^{t_f} \frac{dt}{RC} \Rightarrow$$

$$\ln\left[\frac{\epsilon \cdot C - q(t)}{\epsilon \cdot C}\right] = -\frac{t}{RC} \Rightarrow$$ SIGNIFICA QUE:

$$\frac{\epsilon \cdot C - q(t)}{\epsilon \cdot C} = e^{-t/RC} \Rightarrow \epsilon \cdot C - q(t) = \epsilon \cdot C \cdot e^{-t/RC}$$

O SEA QUE:

$$q(t) = \epsilon \cdot C - \epsilon \cdot C \cdot e^{-t/RC} \Rightarrow$$

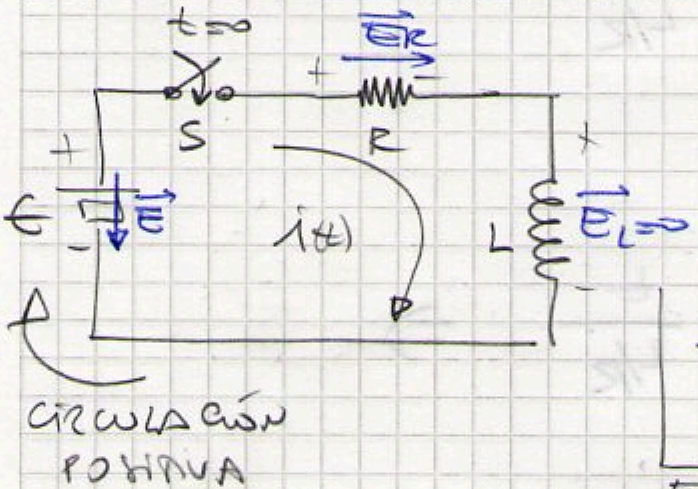
$$\boxed{q(t) = \epsilon \cdot C\left[1 - e^{-t/RC}\right]}$$

PERO COMO: $\epsilon \cdot C \rightarrow Q_{FINAL}$ y ACEPTANDO QUE $\boxed{\tau = RC}$

$$\boxed{q(t) = Q_{FINAL}\left[1 - e^{-t/\tau}\right]}$$ e

$$\boxed{i(t) = \frac{Q}{R \cdot C} \cdot e^{-t/\tau} = I_0 \cdot e^{-t/\tau}}$$

RESOLUCIÓN DEL CIRCUITO R-L =



$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt} = -L \cdot \frac{di(t)}{dt}$$

$$\boxed{-\epsilon + i(t) \cdot R + 0 = -L \cdot \frac{di(t)}{dt}}$$

ESTA ES LA ECUACIÓN

REACOMODANDO = (MUY SIMILAR AL CASO "R-C")

$$\boxed{\frac{di(t)}{dt} + i(t)\frac{R}{L} - \frac{\epsilon}{L} = 0}$$

ECUACIÓN DIFERENCIAL DE PRIMER ORDEN

REACOMODANDO (OTRA VEZ!):

$$\frac{di(t)}{dt} = \frac{R}{L}\left(\frac{\epsilon}{R} - i(t)\right) = \frac{1}{L/R}\left(\frac{\epsilon}{R} - i(t)\right)$$

ASÍ QUE:

$$\int_{i_i}^{i_f} \frac{di}{\left(\frac{\epsilon}{R} - i(t)\right)} = \int_0^{t_f} \frac{dt}{L/R} \Rightarrow$$

CAMBIO DE VARIABLES:

$$\mu = \frac{\epsilon}{R} - i(t)$$

DE TAL MANERA QUE:

$du = -di$ y SUS LÍMITES SERÁN

$$\begin{cases} t=0 \rightarrow i = 0 \rightarrow \mu_i = \frac{\epsilon}{R} \\ t = t_f \rightarrow i = i(t_f) \rightarrow \mu_f = \frac{\epsilon}{R} - i(t) \end{cases}$$

OTRA VEZ =

$$\int_{\frac{\epsilon}{R}}^{\frac{\epsilon}{R}-i(t)} \frac{du}{u} = -\int_{0}^{t_f} \frac{dt}{L/R}$$

$$Ln\left(\frac{\frac{\epsilon}{R}-i(t)}{\frac{\epsilon}{R}}\right) = -\frac{t}{L/R} \Rightarrow$$

$$\frac{\frac{\epsilon}{R}-i(t)}{\frac{\epsilon}{R}} = e^{-t/L/R} \Rightarrow$$

$$\frac{\epsilon}{R} - i(t) = \frac{\epsilon}{R} \cdot e^{-t/L/R} \Rightarrow$$

$$\boxed{i(t) = \frac{\epsilon}{R} \cdot \left[1 - e^{-t/L/R}\right]}$$

PERO COMO $\frac{\epsilon}{R} = I_{MAX}$ y ACEPTANDO QUE $\tau = \frac{L}{R} \Rightarrow$

$$\boxed{i(t) = I_{MAX}\left[1 - e^{-t/\tau}\right]} \quad y$$

$$v_L(t) = L \cdot \frac{di(t)}{dt} = L \cdot \frac{I_{MAX}}{\tau} \cdot e^{-t/\tau} = \underbrace{\left(L \cdot \frac{I_{MAX}}{L} \cdot R\right)}_{\epsilon} \cdot e^{-t/\tau}$$

$$\boxed{v_L(t) = \epsilon \cdot e^{-t/\tau}}$$